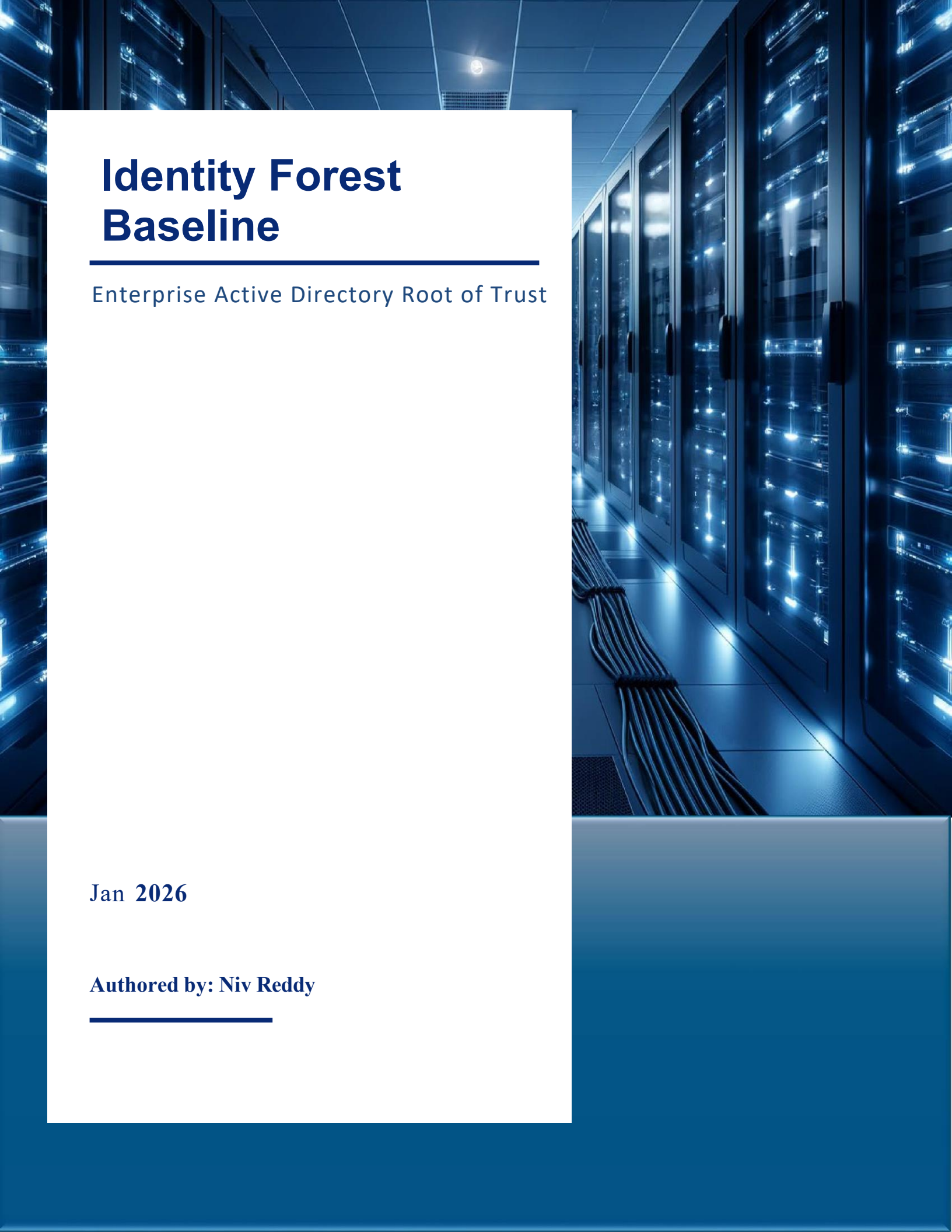




Identity Forest Baseline

Enterprise Active Directory Root of Trust

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Introduction

Context

This document defines the baseline identity architecture of the lab.niv.local enterprise environment.

It records the foundational trust boundary from which all authentication, authorization, security monitoring, and adversary simulation layers will extend.

This forest represents the root of trust for:

- Domain Authentication
- Name Resolution Authority
- Kerberos Ticketing
- Policy Enforcement
- Logging Correlation
- Privilege Modeling
- Security Simulation

All future systems and security controls are layered onto this baseline.

Forest Root Identity Plane

This register defines the canonical identity authority for the lab.niv.local enterprise environment. All authentication, authorization, Kerberos ticketing, DNS resolution, logging correlation, privilege modeling, and security simulation layers anchor to this platform.

Forest and Domain Authority

Component	Value
Forest Root	lab.niv.local
Forest Mode	Windows Server 2022
Domain Mode	Windows Server 2022
Domain Controller	DC01
Directory Services	AD DS, DNS, Kerberos, LDAP

Domain Controller Authority Register

Attribute	Value
Hostname	DC01
OS	Windows Server 2022
IP Address	10.0.10.10
Role	Forest root domain controller
DNS Role	Internal authoritative DNS
FSMO Roles	All FSMO roles hosted locally

Platform Fabric Construction

Purpose

Before any identity services were deployed, a sealed corporate network fabric was engineered at the hypervisor layer to serve as the physical trust boundary of the identity platform.

Fabric Implementation

A dedicated internal Layer-2 Linux bridge (vmbr1) was created on the Proxmox host

```
auto vmbr1
iface vmbr1 inet manual
    bridge-ports none
    bridge-stp off
    bridge-fd 0
```

Resulting Trust Boundary

This fabric operates as a virtual enterprise switch that provides:

- A private corporate broadcast domain
- No routing or internet connectivity
- No residential DNS or DHCP exposure
- A sealed environment for security simulation

All identity systems attach exclusively to this fabric.

Domain Controller Virtualization Architecture (DC01)

Purpose

DC01 was intentionally engineered as a modern, hypervisor-aware Windows Server 2022 guest to ensure deterministic I/O, predictable authentication behavior, and stable long-term operation

Virtual Machine Architecture

Component	Configuration
Firmware	UEFI (OVMF)
Machine Type	q35
Storage Controller	VirtIO SCSI
Disk	60 GB
Disk Mode	Write-back cache, async I/O
Network	VirtIO on vmbr1

Paravirtualization Controls

VirtIO drivers were injected during OS installation to ensure native hypervisor integration before any directory services were deployed.

Design Intent

DC01 was treated as infrastructure, not a generic VM. It was engineered as the authoritative identity root with deterministic storage, sealed networking, and no dependency on residential services

Domain Controller Pre-Promotion Hardening

Network Identity Control

DC01 was statically assigned **10.0.10.10/24** and configured to use itself as its authoritative resolver.

Directory Services Staging

AD DS and DNS Server roles were installed and staged for promotion.

Forest and Domain Authority Establishment

DC01 was promoted to forest root domain controller, and a new forest was created:

Forest root: **lab.niv.local**

This initialized:

- AD DS
- DNS
- Kerberos
- Global Catalog
- SYSVOL
- NTDS

DC01 became the authentication authority, DNS owner, Kerberos KDC, and trust anchor.

Corporate DNS Authority Hardening

- Secure dynamic updates only
- Reverse zone authority created
- IPv6 removed
- DNS bound only to internal interface
- Forward and reverse resolution validated

Kerberos Realm Validation

- TGT issuance validated (AES-256)
- Realm binding confirmed
- Domain controller discoverability verified
- Time authority validated

Identity Structure and Administrative Tiering

Core OUs:

- **Servers**
- **Workstations**
- **Admins**
- **Service Accounts**
- **Security Groups**

Administrative Tiering:

Tier	Purpose
Tier 0	Forest-level control (schema, domain, DC administration)
Tier 1	Server-level administrative control
Tier 2	Workstation and user-level control

Tiered Security Control Groups

Group	Function
T0-Domain-Admins	Domain administration
T0-Schema-Admins	Schema modification
T0-Enterprise-Admins	Forest-wide control
T0-DC-Admins	Domain controller local admin
T1-Server-Admins	Member server administrators
T1-Server-Operators	Limited server operations
T1-Service-Accounts	Server service identities
T2-Workstation-Admins	Workstation administrators
T2-Helpdesk	Password resets & unlock
T2-Standard-Users	Normal domain users

Identity Governance and Change-Control Baseline

Naming Authority Standards:

Object Type	Naming Standard	Purpose
Standard users	first.last	Human identity namespace
Tier admin users	first.last-t0 first.last-t1 first.last-t2	Privileged separation
Workstations	WS-LOCATION-##	Physical device attribution
Servers	SRV-ROLE-##	Functional role attribution
Service accounts	svc-application	Non-interactive service identity
Tier security groups	T0-* T1-* T2-*	Privilege control planes
Domain local groups	DL-*	Permission assignment layer
Global identity groups	GG-*	Role aggregation layer